

École Nationale Supérieure de l'Électronique et de ses Applications

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Project: GenAI for molecular discovery

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Abstract. *This report investigates the application of advanced graph-based deep learning architectures to predict the olfactory properties of molecules directly from their chemical structures. Utilizing a dataset combining the GoodScents and Leffingwell databases, the task is formulated as a multi-label classification problem across 138 distinct odor descriptors. Due to the dataset's extreme class imbalance and zero-inflation, classical machine learning approaches and naive neural network implementations fail to achieve robust predictive utility. To address this, we represented molecules as topological graphs and evaluated a series of Graph Neural Networks (GNNs), including Graph Convolutional Networks (GCNs), Graph Attention Networks v2 (GATv2), and Message Passing Neural Networks (MPNNs). Furthermore, we introduced a rigorous methodology utilizing second-order iterative stratification for representative data splitting and independent per-label threshold optimization to maximize the Micro-F1 score. Our results demonstrate that this fully optimized pipeline significantly enhances predictive performance, with our MPNN architecture achieving an AUROC of 0.9001 and an F1-Score of 0.3895, successfully surpassing published benchmark metrics. Ultimately, this work confirms that deep graph representations, when coupled with strict threshold calibration and stratified sampling, effectively bridge the gap between chemical topology and biological scent perception.*

1. Introduction

Research into the relationship between molecular structure and perceived odor has the objective of demystifying a highly complex biological mechanism. Discoveries in this field significantly impact the synthetic fragrance industry, which is crucial for food, beverages and cosmetics, while also advancing our understanding of sensory neuroscience. Improvements in the manufacturing of synthetic odorants can help conserve natural resources and optimize time-consuming laboratory processes. In this context, machine learning is increasingly being explored for olfaction, mirroring its transformative success in the senses of sight and hearing. The literature demonstrates that models trained to predict the olfactory qualities of molecules can, in some cases, rival or exceed human expert consistency. A robust predictive system has profound practical applications: recognizing the mapping between a molecule's structure and its odor enables the targeted creation of artificial scents, benefiting chemistry, biology, and medicine.

This report describes our continued work on the "GenAI for Molecular Discovery" project, carried out under the supervision of Professor G. Renton. The core objective of this project is to explore how to identify and predict the odor of a molecule simply by analyzing its chemical structure. In practice, our work focuses on constructing an advanced Deep Learning model that, given a large dataset of molecules, can correctly predict the complex odor profile associated with each molecule.

In recent years, the field of olfactory informatics has transitioned away from traditional feature engineering toward advanced deep learning architectures capable of modeling the complex Structure-Odor Relationship (SOR). Notably, foundational work has established the efficacy of Graph Neural Networks (GNNs) [2], demonstrating that treating molecules directly as learnable graphs—where atoms are nodes and bonds are edges—captures highly relational structural information and yields state-of-the-art ranking performance for multi-label odor prediction. Expanding on these deep learning capabilities, complementary multimodal approaches have explored integrating physicochemical properties with computer vision techniques, ensembling Deep Neural Networks (DNNs) with Convolutional Neural Networks (CNNs) trained on

2D molecular images to achieve exceptionally high classification accuracy [3]. Furthermore, recent extensions of graph-based models [1] have incorporated a biological perspective into the SOR by successfully predicting both broad odor categories and compound-receptor interactions. Together, these advancements reveal that modern machine learning can effectively map olfactory structures through diverse, highly expressive representations, successfully bridging the gap between chemical topology and biological scent perception.

Our study involves the analysis of a combined dataset from GoodScents and Leffingwell, encompassing a vast list of molecules categorized across 138 distinct odor descriptors. Because molecules typically possess only a few distinct odors, the dataset presents a severe extreme class imbalance (zero-inflation) challenge. To address this, our methodology shifts entirely to state-of-the-art graph-based deep learning. We aim to construct a sophisticated GNN capable of achieving excellent results in this extreme multi-label classification task. By applying advanced dataset splitting techniques, such as second-order iterative stratification, and fine-tuning network hyperparameters, we aim to extract the most relevant topological features from the data to achieve results comparable to the current state-of-the-art.

To achieve this, the project builds a predictive system directly from SMILES representations, converting them into graph structures to leverage message-passing neural networks (like GCNs). To improve our model's predictive power and provide comprehensive node features, additional chemical properties are extracted using RDKit and integrated into the graph.

Finally, our goals include rigorously evaluating the model using metrics most relevant to the sparse nature of olfactory machine learning literature. Moving beyond standard accuracy, we prioritize the Area Under the Receiver Operating Characteristic (AUROC) and the Precision-Recall Curve (AUPRC) to measure ranking capability. Furthermore, we implement sophisticated post-processing techniques, such as per-label threshold optimization, to maximize the F1-score, ensuring our model's predictions are both sensitive and precise for practical, real-world application.

The remainder of this report is organized as follows: In Section 2, we review the current State-of-the-Art, exploring recent advancements in olfactory informatics, including graph-based deep learning and multimodal architectures. Section 3 details the Dataset, discussing its chemical origins, the inherent challenge of extreme class imbalance, and our implementation of second-order iterative stratification. In Section 4, we outline our Methodology, detailing the graph-based data preprocessing, the theoretical foundations of the employed neural architectures (GCN, GATv2, and MPNN), the advanced training setups, and our evaluation metrics. Section 5 presents our Results, analyzing the performance of our baseline models and demonstrating the critical impact of our optimized data handling pipeline in replicating and exceeding literature benchmarks. Section 6 provides the Conclusion, summarizing our key findings and the overall impact of the project. Finally, Section 7 discusses Future Works, highlighting possible improvements to be done with this work.

2. State-of-the-art

The field of olfactory informatics has undergone a significant transformation in recent years, evolving from traditional approaches based on handcrafted molecular descriptors toward more advanced deep learning methodologies capable of modeling the complex Structure-Odor Relationship (SOR). Within this context, recent literature highlights a progressive shift toward representations that better capture molecular structure and perceptual relationships, particularly through graph-based learning and multimodal deep learning strategies. Below, we review key contributions that have shaped the current state of the art in odor prediction from molecular

data.

A foundational line of work proposes the use of Graph Neural Networks (GNNs) for Quantitative Structure-Odor Relationship (QSOR) modeling [2], where molecules are directly represented as graphs with atoms as nodes and bonds as edges. In this framework, a dataset constructed from the combination of the GoodScents and Leffingwell databases includes approximately 5,030 molecules annotated with 138 odor descriptors after standardization and filtering of rare labels. The problem is formulated as a multi-label classification task, reflecting the fact that each molecule may be associated with multiple odor percepts simultaneously, with strong class imbalance and label co-occurrence patterns. In contrast to traditional approaches relying on engineered molecular features such as Morgan or path-based fingerprints generated with RDKit, this methodology employs message-passing neural networks to learn representations directly from graph structure [2]. The architecture consists of iterative neighborhood aggregation followed by fully connected layers, producing both prediction outputs and intermediate embeddings that encode perceptual similarities between molecules. Baseline comparisons include Random Forests and k-NN models trained on fingerprint-based representations. Experimental results show that GNNs achieve state-of-the-art performance, with an AUROC of approximately 0.894 [2], outperforming classical models such as Random Forests on Mordred descriptors, which achieve around 0.850. Furthermore, the learned embeddings demonstrate strong transferability, achieving competitive performance on external benchmarks such as the DREAM Olfaction Challenge. These results indicate that graph-based representations are capable of capturing both structural and perceptual relationships in a unified manner, while also leveraging multi-task learning across odor labels and implicitly modeling their co-occurrence structure.

Another important research direction explores deep learning models built directly from SMILES-derived representations using complementary modalities [3]. In this context, a dataset of 5,185 chemical compounds annotated with 104 odor percepts is used to train two distinct models based on different feature extraction strategies. The first approach relies on physicochemical properties and molecular fingerprints (PPMF) derived from SMILES strings, which are used as input to a Deep Neural Network (DNN). The second approach converts SMILES strings into 2D molecular images, which are then processed using a Convolutional Neural Network (CNN) based on the Xception architecture. Both models are trained as multi-label classifiers with sigmoid outputs, and their predictions are combined using an ensemble strategy based on majority voting. Evaluation is performed using multiple metrics, including accuracy, AUROC, precision, recall, and F1-score. The results indicate very high predictive performance, with the CNN achieving 98.3% accuracy and the DNN achieving 97.3% on an independent test set. The ensemble further improves robustness, achieving over 90% precision for 102 out of 104 odor classes and perfect precision for 64 classes. Despite these strong classification results, AUROC scores remain lower than those reported in graph-based approaches, suggesting differences in ranking performance. An important observation is that the two models exhibit complementary strengths, performing better on different subsets of odor classes, which highlights the relevance of multimodal representations. This line of work emphasizes the effectiveness of engineered features and image-based representations derived from SMILES, while not explicitly modeling molecular graph topology or leveraging learned structural embeddings.

A complementary extension of graph-based approaches introduces Graph Convolutional Networks (GCNs) [1] for both odor prediction and olfactory receptor interaction modeling, thereby incorporating a biological perspective into the structure-odor relationship. In this setting, molecular structures are again represented as graphs, and GCN architectures are

used to learn node-level features that are aggregated into molecule-level representations. The study considers datasets where odor descriptors are grouped into broader categories [1], such as 23 odor classes, and also evaluates compound-receptor interactions. Performance is assessed using metrics such as AUROC and PRC-AUC, with the GCN achieving strong results, including AUROC values around 0.92 and PRC-AUC around 0.67 for odor prediction tasks, significantly outperforming traditional models based on Morgan fingerprints and RDKit descriptors. For receptor prediction, performance is even higher, with AUROC values approaching 0.99 [1], reflecting the ability of graph-based models to capture molecular interaction patterns. Comparisons with benchmarks such as the DREAM Olfaction Challenge further confirm the competitiveness of this approach. However, results vary across odor categories, with better performance observed for well-represented classes and lower performance for rarer or more ambiguous descriptors. This work highlights the advantage of integrating chemical and biological information [1], as well as the impact of working with grouped odor categories rather than fine-grained descriptor spaces, providing additional interpretability while maintaining strong predictive performance.

A recent line of work further revisits the effectiveness of traditional machine learning models when combined with carefully engineered molecular fingerprint representations, providing an important comparative perspective to deep learning approaches. In this study, a large-scale dataset comprising 8,681 odorant molecules aggregated from ten expert-curated sources is constructed and standardized into a multi-label classification framework with over 200 odor descriptors [4]. The methodology systematically benchmarks different feature representations, including functional group fingerprints, physicochemical descriptors, and Morgan structural fingerprints, across tree-based models such as Random Forest, XGBoost, and LightGBM . Results indicate that models based on Morgan fingerprints consistently outperform descriptor-based approaches, with the XGBoost model achieving an AUROC of approximately 0.828 and demonstrating strong robustness under cross-validation [4]. Beyond predictive performance, the study highlights the ability of fingerprint-based models to capture meaningful structural motifs and reconstruct a continuous and interpretable odor space, revealing perceptual relationships between odor categories through embedding and dimensionality reduction techniques . These findings emphasize that, despite the growing dominance of deep learning architectures, well-designed feature representations combined with classical machine learning algorithms remain highly competitive, offering advantages in interpretability, computational efficiency, and scalability for large, heterogeneous olfactory datasets.

Taken together, these works illustrate the diversity of methodological approaches currently employed in olfactory prediction tasks, ranging from graph-based deep learning models that operate directly on molecular topology to multimodal architectures leveraging engineered descriptors and image representations derived from SMILES. They also reveal important variations in label granularity, evaluation metrics, and strategies for handling class imbalance, as well as differences in how structural and perceptual relationships are modeled. This body of research collectively demonstrates that while graph neural networks provide a powerful framework for learning rich molecular representations, alternative approaches based on complementary feature spaces can achieve strong performance under different evaluation criteria, contributing to a broader understanding of the structure-odor relationship.

3. Dataset

The dataset utilized for this project is a synthesis of two prominent sources: The Good Scents Company Information System and the Leffingwell PMP 2001 database. The Good Scents database supplies qualitative odor descriptions widely employed in flavor and fragrance

chemistry, curated through expert sensory evaluations and professional literature. Complementing this, the Leffingwell dataset provides computationally standardized odor descriptors categorized by industry experts. Combining these resources yields a robust, professional-grade multi-label dataset where each molecule is mapped to one or more subjective, descriptive olfactory labels.

3.1. Dataset Characteristics & Extreme Class Imbalance

The combined dataset defines a comprehensive taxonomy of 138 distinct odor descriptors. A defining and highly challenging characteristic of this chemical data is severe "zero-inflation," or extreme class imbalance. Because an average molecule possesses only a narrow profile of distinct odors—typically 3 to 4 out of the possible 138—approximately 97% of the label matrix consists of negative instances (zeros).

Furthermore, the base frequency of individual labels varies drastically. While ubiquitous labels or lack thereof (such as "odorless," which appears at a 4% rate) are relatively common, highly specific or highly subjective labels are exceptionally rare. This structural sparsity necessitates specialized modeling and evaluation techniques, as conventional machine learning algorithms will naturally bias toward predicting the absence of an odor to artificially minimize global loss, resulting in high accuracy but zero practical predictive power for the actual scents.

3.2. Data Splitting: Second-Order Iterative Stratification

The multi-label nature and extreme imbalance of the 138 descriptors severely complicate the creation of training, validation, and test splits. Standard randomized splitting, such as the 80/20 split used in our preliminary experiments, is inadequate for this phase of research. In a highly imbalanced multi-label scenario, isolating a molecule to satisfy the training distribution of one specific label inadvertently disrupts the distributions of its co-occurring labels.

To resolve this and ensure reliable evaluation metrics, we implemented second-order iterative stratification. Unlike standard stratification, this algorithm explicitly preserves the ratios of label pairs (co-occurrences) across all data splits. By prioritizing the distribution of the rarest label combinations first, we mathematically guarantee that our validation and test sets are highly representative, miniature versions of the overall dataset. This prevents ultra-rare molecular scent profiles from being completely excluded from the evaluation phase, allowing for accurate tuning of decision thresholds and robust calculation of Precision, Recall, and F1-scores.

4. Methodology

4.1. Dataset preprocessing

In the dataset utilized for our project, each sample is initially defined by the Simplified Molecular Input Line Entry System (SMILES) representation of a molecule, coupled with a set of associated odor descriptors. From this raw data, a global set of unique odor labels was constructed, resulting in a complex multi-label classification problem with 138 distinct output labels.

Rather than relying on classical feature engineering or simple atom-count matrix representations, our methodology required transforming the data into a format natively suited for Graph Neural Networks (GNNs). Using cheminformatics toolkits such as RDKit, we converted each SMILES string directly into a topological graph structure. In this formulation, atoms are represented as nodes, and the chemical bonds connecting them are represented as edges (see

Figure 1). To enrich the predictive power of the model, we extracted relevant physicochemical properties to serve as node features (e.g., atom type, hybridization state, and partial charge) and edge features (e.g., bond type). This graph-based representation allows the neural network to learn complex structural, spatial, and relational chemical information directly from the molecule's topology.

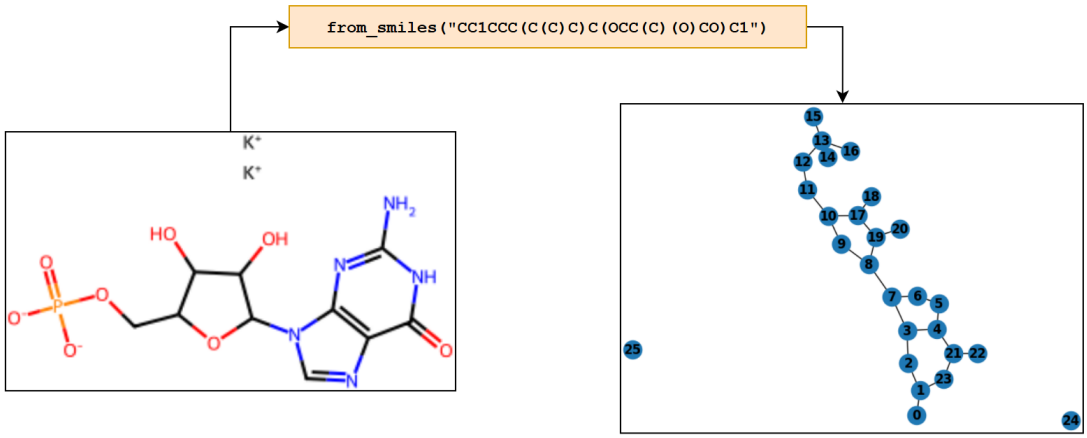


Figure 1. Example of a conversion of a molecule from SMILES to the graph representation.

The target output for each molecular graph is a binary vector of length 138, indicating the presence (1) or absence (0) of each specific odor descriptor. As observed during the label extraction process, the dataset exhibits a severe class imbalance—or zero-inflation—which is a common characteristic of olfactory datasets. The vast majority of the target matrix consists of zeros, as most molecules are associated with only a handful of scents.

Because of this extreme sparsity and the highly correlated nature of olfactory labels, standard randomized splitting methods (such as a basic 80/20 random split) are entirely inadequate. Isolating molecules to satisfy the training distribution of one label often disrupts the natural frequency of co-occurring labels in the test set. To resolve this, the dataset was partitioned into training, validation, and test sets using second-order iterative stratification. This advanced algorithm systematically preserves the ratios of label pairs (co-occurrences) across all splits. By explicitly prioritizing and distributing the rarest combinations of odors proportionally, we mathematically guarantee that our validation and test sets remain highly representative miniature versions of the overall dataset. This sophisticated splitting approach is critical for enabling robust hyperparameter tuning and ensuring rigorous, unbiased final model evaluation. Apart from that, the chosen batch size was 32.

4.2. Graph Neural Networks (GNNs)

After completing the data preprocessing and converting the molecular SMILES into graph structures, we shifted our focus to training deep learning models capable of operating natively on these graphs. Given that each molecule can be associated with multiple odor descriptors simultaneously, the problem was formulated as a multi-label classification task. To address this, we transitioned exclusively to Graph Neural Networks (GNNs), which represent the current state-of-the-art for mapping molecular structure to olfactory perception.

4.2.1. Graph Neural Networks (GNNs): Theoretical Overview

Graph Neural Networks are a class of deep learning architectures designed to perform inference on data described by graphs. Unlike standard neural networks that expect fixed-size vectors, GNNs can process molecules of varying sizes and shapes. They achieve this through a process called "message passing." During each forward pass, every atom (node) aggregates feature information from its immediate neighbors. By stacking multiple message-passing layers, the network allows information to flow further across the molecule, enabling each atom to understand its local chemical environment within the broader molecular context.

After the message-passing phase, a "readout" or pooling function (such as sum, mean, or max) is applied to aggregate all individual node representations into a single, fixed-length vector that represents the entire molecule. This comprehensive molecular embedding is then passed through a standard fully connected Multi-Layer Perceptron (MLP) to generate the final predictions.

4.2.2. Baseline GNN Exploration

Before implementing complex architectures inspired by the literature, we established a baseline using a standard Graph Convolutional Network (GCN). This initial phase was highly empirical, focused on understanding how different structural hyperparameters influenced the model's ability to capture odor-related patterns.

The core architecture for these baseline experiments relied on sequential message-passing layers utilizing the standard `GCNConv` operator, with Rectified Linear Unit (ReLU) activations applied after each convolution. We primarily explored networks with varying depths—specifically 2, 3, and 4 layers. The 4-layer architecture progressively expanded and then compressed the hidden feature dimensions, utilizing sequential layer sizes of 64, 128, 256, and 128. Following this message-passing phase, the node embeddings were aggregated using a global pooling function, and the resulting 128-dimensional graph representation was passed through a single linear classification layer to predict the 138 odor classes. The shallower variations followed this exact structural foundation but truncated the depth; for example, the 2-layer variant utilized only the first two convolutional layers (sizes 64 and 128) before proceeding directly to the pooling and linear classification steps.

During this exploration, we observed that too few layers prevented the model from capturing complex functional groups, while too many layers risked "oversmoothing," where the representations of different atoms became indistinguishable. Alongside modifying the depth, we also experimented with different readout functions, specifically swapping `global_mean_pool` for `global_max_pool`. While mean pooling averages the features of all atoms (capturing global molecular properties), max pooling isolates the most prominent atomic features. This distinction is often critical in olfaction, where the presence of a single, highly reactive functional group can dominate a molecule's overall scent profile.

4.2.3. Literature-Approximated GCN Architecture

To elevate our predictive performance and achieve results comparable to published benchmarks, we constructed a new GCN architecture meticulously designed to approximate the leading model described in the literature (specifically referencing the optimized GNN from

Sanchez-Lengeling et al.).

This advanced architecture incorporates several specific structural and hyperparameter choices tailored to the extreme complexity of the 138-label olfactory task:

- **Message Passing Layers:** The model utilizes 4 sequential GCN layers with progressively increasing hidden dimensions of [15, 20, 27, 36]. Instead of standard ReLU activations, this stage utilizes Scaled Exponential Linear Units (SELU) to induce self-normalizing properties during the message-passing phase.
- **Readout Strategy (Jumping Knowledge):** Rather than only pooling the final layer, the model utilizes a concatenated global sum pooling approach. The graph representations from the input features and every intermediate message-passing layer are individually sum-pooled and then concatenated. This creates a rich, $\sim$175-dimensional molecular embedding that retains both local, low-level atomic details and broad, high-level structural patterns.
- **Fully-Connected Neural Net:** The molecular embedding is passed through a 2-layer MLP with dimensions [96, 63]. To prevent overfitting on our dataset, aggressive regularization is applied, incorporating Batch Normalization and a high Dropout rate of 0.47 after each layer, alongside ReLU activations.
- **Prediction Head:** The final layer is a linear transformation mapping the 63-dimensional vector to the 138 multi-label tasks.

4.2.4. Graph Attention Network V2 (GATv2)

After establishing a strong baseline using Graph Convolutional Networks and exploring their limitations in capturing complex molecular interactions, we shifted our focus toward more expressive architectures capable of dynamically weighting the importance of atomic neighborhoods. Given the inherently heterogeneous nature of molecular graphs—where certain atoms or functional groups can disproportionately influence olfactory perception—we introduced attention-based mechanisms into our modeling pipeline. This led to the adoption of Graph Attention Networks (GATs) and their improved variant, GATv2.

Theoretical Overview: GATs extend the standard GNN framework by incorporating an attention mechanism. Unlike GCNs, where neighbor contributions are aggregated using fixed or degree-normalized weights, GATs allow the model to learn adaptive, data-dependent weights for each edge in the graph. During each step, a node computes attention coefficients for all its neighbors based on their feature representations. These coefficients are normalized using a softmax function, and the node's updated representation is computed as a weighted sum of its neighbors' features.

Motivation for GATv2: Standard GAT models suffer from a structural limitation where the relative importance assigned to neighboring nodes may not fully depend on the specific query node. GATv2 addresses this by applying linear transformations *after* concatenating the features of the source and target nodes, enabling fully dynamic attention ranking.

Architecture Implementation: The implemented GATv2 architecture consists of two sequential GATv2 convolutional layers. The first layer employs multi-head attention, concatenating the outputs and passing them through an Exponential Linear Unit (ELU) activation function and dropout. The second GATv2 layer uses a single attention head with concatenation disabled. Following the message-passing phase, a global mean pooling operation aggregates node-level representations into a single graph-level embedding, which is finally mapped to the target space via a fully connected linear layer.

4.2.5. Message Passing Neural Network (MPNN)

We also tested a Message Passing Neural Network (MPNN), which operates by propagating and aggregating information across nodes and edges, allowing nodes to capture structural information from increasingly distant parts of the graph. Our MPNN was implemented with the specific architectural hyperparameters developed by Sanchez-Lengeling et al. [2]. The model features five message passing layers with a hidden dimension of 43, employing an edge-conditioned matrix multiplication message type and a GRU-based update function. The readout phase utilizes global sum pooling with a 197-dimensional softmax, incorporating residual connections, followed by a 3-layer fully-connected network with 392 units, ReLU activations, batch normalization, and a dropout rate of 0.12.

4.2.6. Advanced Training Setup

Because our dataset exhibits extreme class imbalance, standard loss functions fail to optimize the model effectively. To counter this, the network was trained using a Weighted Binary Cross-Entropy Loss (`BCEWithLogitsLoss`). We dynamically calculated a positive weight (`pos_weight`) vector for all 138 classes based on their frequency in the training set, heavily penalizing the model for missing rare odor descriptors (False Negatives).

The models were optimized using the Adam optimizer with an initial learning rate of 0.001. For the training of our Base GCN architectures, we employed an early stopping regularization method with a patience of 50 epochs; this halted the training process if validation metrics ceased to improve. In contrast, to navigate the complex loss landscape of the advanced architectures, we implemented a Cosine Annealing learning rate scheduler with Warm Restarts over a fixed run of 300 training epochs, helping the deeper models escape local minima.

4.3. Evaluation metrics

To rigorously evaluate the performance of our models on a severely imbalanced multi-label classification task, standard single-label metrics like simple accuracy are insufficient. Instead, we focused on metrics that capture the trade-offs between false positives and false negatives.

4.3.1. Precision and Recall

Precision measures the proportion of true positive predictions out of all positive predictions made by the model, indicating its reliability:

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

Recall (Sensitivity) measures the proportion of actual true positives that the model successfully identified:

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

4.3.2. F1-Score and Threshold Optimization

The F1-score is the harmonic mean of Precision and Recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{3}$$

Due to extreme zero-inflation, evaluating raw probabilities at a default threshold of 0.5 yields heavily skewed utility results. We implemented a rigorous threshold optimization strategy, sweeping through all possible decision thresholds on our validation set to find the exact cutoffs that maximized the F1-score. This optimized threshold was then applied during final test set evaluation.

4.3.3. AUROC

The Area Under the Receiver Operating Characteristic Curve (AUROC) evaluates the model's ability to correctly rank positive instances higher than negative ones, independent of any specific decision threshold. An AUROC score of 1.0 indicates perfect separation between classes.

4.3.4. Auxiliary Metrics for Literature Comparison

While Precision, Recall, F1-Score, and AUROC form the core of our evaluation methodology, we may introduce supplementary metrics (such as Hamming Loss) when strictly necessary to ensure direct, one-to-one comparisons with benchmark models in published state-of-the-art articles.

5. Results

5.1. Baseline GCN Experiments: Readout Function and Architecture Depth

The initial phase of our results analysis focused on evaluating the baseline Graph Convolutional Network (GCN) architecture. We systematically explored the impact of two fundamental structural hyperparameters: the choice of readout function (`global_mean_pool` versus `global_max_pool`) and the network depth (number of message-passing layers), while maintaining a fixed learning rate ($LR = 0.001$) over 100 training epochs. All metrics reported in this section were calculated on the dedicated test set using a default classification threshold of 0.50. The quantitative results are summarized in Table 1.

Readout function	No. of Layers	Hamming Loss	Micro F1-Score	Subset Accuracy
Global mean pool	2	0.1907	0.1291	0
	3	0.1815	0.1436	0.0410
	4	**0.1080**	**0.1954**	**0.0641**
Global max pool	2	0.1616	0.1600	0.0401
	3	0.1023	0.1915	0.0561
	4	**0.0784**	**0.2259**	**0.0641**

Table 1. Baseline GCN performance metrics.

5.1.1. Quantitative Performance Analysis

Analysis of the test set evaluation reveals distinct performance trends correlated with architectural choices. Our primary finding is that GCNs employing `global_max_pool` consistently outperformed their `global_mean_pool` counterparts across all key metrics when layer counts were held constant. Holding the architecture fixed at two message-passing layers, swapping the readout function raised the Micro-F1 score from 0.1291 to 0.1600. This differential became more pronounced at a depth of four layers, where the Max Pooling model achieved an F1-score of 0.2259—the highest score in this baseline group—compared to 0.1954 for the equivalent Mean Pooling model.

Furthermore, increasing the network depth yielded performance gains for both readout strategies. For Mean Pooling architectures, increasing from 2 layers to 4 layers produced a monotonic improvement, confirming that multi-layer message passing is necessary to capture complex, multi-hop chemical relationships.

5.1.2. Readout Function and Training Stability

While prioritizing `global_max_pool` based purely on final metric scores seems intuitive, a qualitative analysis of the training progression reveals a critical trade-off in training stability.

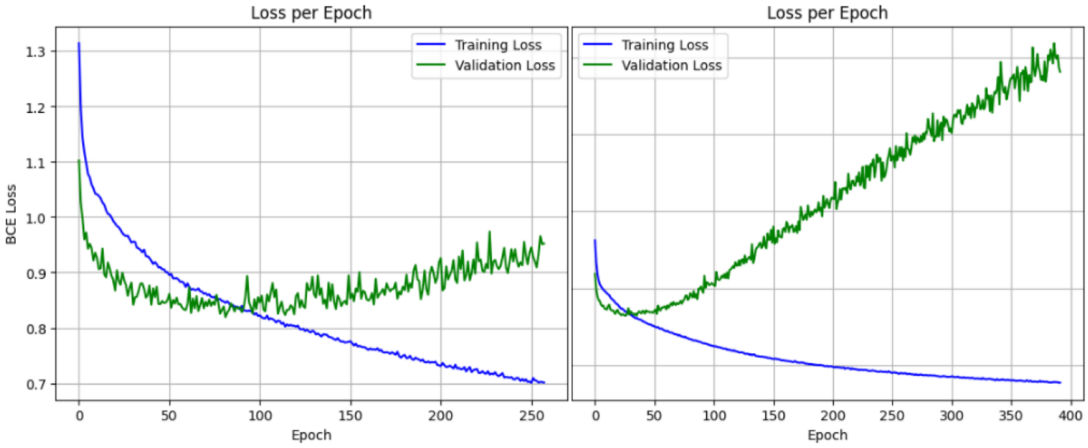


Figure 2. Training and Validation Loss for Mean Pooling (left) and Max Pooling (right).

Architectures utilizing `global_mean_pool` typically generated smooth, monotonically decreasing loss curves, indicative of a highly stable and convergent training process. In stark contrast, models employing `global_max_pool` exhibited significantly more erratic validation loss behavior. Although the general trend was downward, the validation loss was characterized by frequent, large fluctuations. In summary, deeper GCNs with Max Pooling maximize final quantitative performance but introduce severe instability into the training dynamics. Mean pooling provides far more reliable convergence at the cost of peak predictive scores.

5.2. Literature Replication and Advanced Architectures

Our subsequent experiments aimed to replicate the state-of-the-art architectures described in the literature—specifically the optimized GCN and MPNN. We evaluated the critical impact of our advanced data handling techniques (second-order iterative stratification and

threshold optimization) by comparing a "naive" replication against a fully optimized pipeline. Finally, we extended the literature's framework by evaluating a Graph Attention Network v2 (GATv2). The quantitative results are summarized in Table 2.

Model	AUROC	Precision	Recall	F1-Score
GCN (Paper)	0.8900	0.3790	0.3870	0.3620
GCN (Ours)	0.8464	0.1008	0.8109	0.1793
GCN (Ours + Stratification + Thresholds)	0.8561	0.2616	0.5158	0.3471
MPNN (Paper)	0.8940	0.3790	0.3900	0.3600
MPNN (Ours)	0.8642	0.1216	0.6023	0.2023
MPNN (Ours + Stratification + Thresholds)	**0.9001**	0.3090	0.5270	**0.3895**
GATv2	0.8310	0.0713	0.7128	0.1296
GATv2 (Stratification + Thresholds)	0.8561	0.2436	0.5042	0.3284

Table 2. Comparison of advanced architectures and the impact of our methodology pipeline.

5.2.1. GCN Performance and Methodology Impact

Our initial attempts to replicate the literature's GCN model without specialized data splitting or post-processing highlighted the severe challenges of extreme class imbalance. As shown in Table 2, the "naive" implementation evaluated at a default 0.5 decision threshold fell significantly short of published benchmarks, yielding a Micro F1-score of just 0.1793 due to an overwhelming number of false positives diluting precision.

Furthermore, analyzing the training dynamics reveals critical insights into the model's fitting behavior. When trained naively without iterative stratification, the validation loss curve exhibited clear separation from the training loss curve (see Figure 3). This divergence indicates that the model was struggling to generalize, suffering from overfitting as it attempted to navigate the highly skewed, unstratified training distribution.

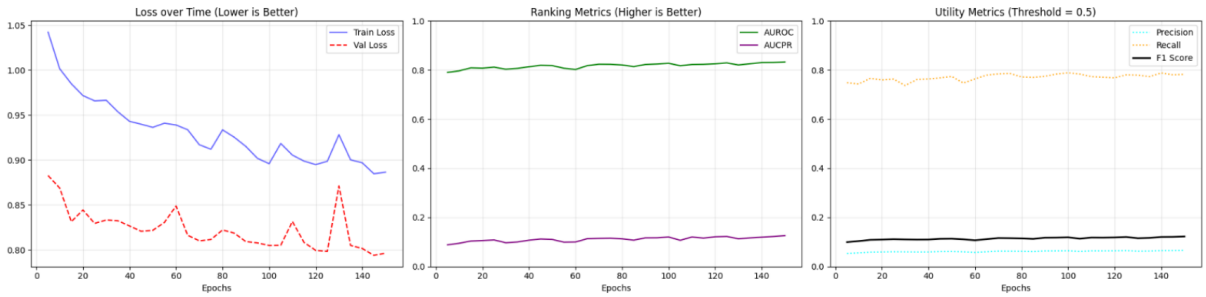


Figure 3. Training and validation loss curves for the naive GCN model without stratification.

However, the application of our advanced methodology drastically transformed both the stability and the utility of this architecture. By integrating second-order iterative stratification, the validation and training loss curves became tightly aligned and tracked each other consistently throughout the training process (see Figure 4). This alignment visually demonstrates that the stratified splitting successfully mitigated the previous fitting problems, ensuring the model learned a robust and generalized representation of the structure-odor relationship.

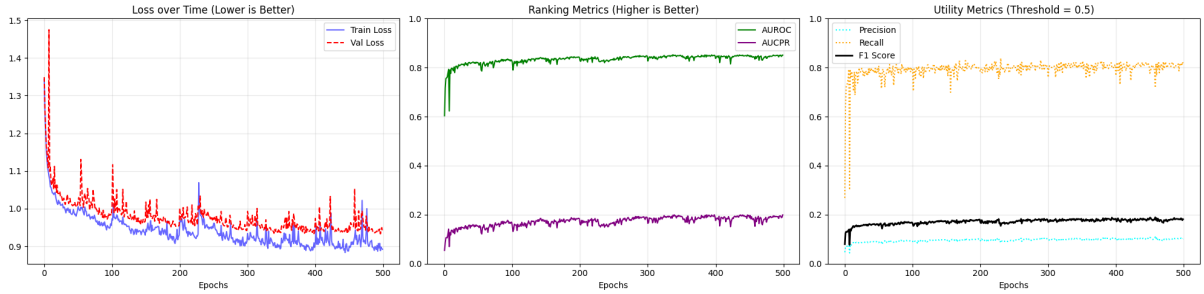


Figure 4. Training and validation loss curves for the fully optimized GCN model with stratification, extended to 500 epochs.

Additionally, monitoring these synchronized loss curves informed a necessary adjustment to our training duration. We observed that at our original target of 300 epochs, the loss was still steadily decreasing, indicating that the network had not yet reached its full potential. Consequently, we extended the training regime for this final optimized GCN model to 500 epochs. This extension provided the architecture adequate time to fully converge, allowing the loss curves to properly flatten out.

Ultimately, pairing this stabilized, extended training with independently optimized decision thresholds allowed the fully optimized GCN pipeline to achieve an AUROC of 0.8561 and an F1-Score of 0.3471, closely bridging the gap with the original paper's reported F1-Score of 0.362.

5.3. MPNN Experiments

The results for the Message Passing Neural Network (MPNN) were even more remarkable and demonstrate the absolute necessity of our data handling pipeline. Initially, the model without these techniques yielded an AUROC of 0.8642 and an F1-Score of 0.2023, where a low Precision of 0.1216 indicated a high rate of false positives.

By introducing stratification and thresholding, the AUROC increased to 0.9001, which is within the range of the state-of-the-art mean of 0.890, and the F1-Score improved to 0.3895, successfully surpassing the original paper's reported 0.362. While our final Precision of 0.3090 remains slightly lower than the 0.379 achieved in the literature, our Recall of 0.5270 is significantly higher than their 0.387, showing that our model identifies a much larger portion of positive cases.

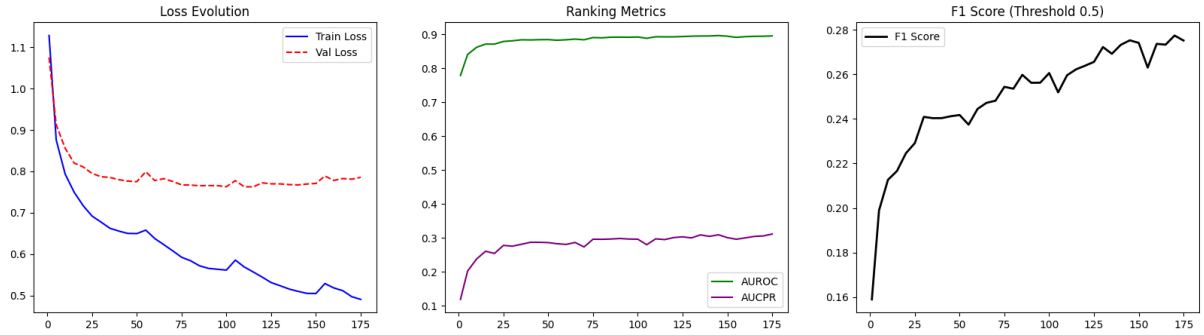


Figure 5. Training dynamics, ranking metrics, and naive F1-score evolution for the MPNN.

The true value of our methodology is fully contextualized by analyzing the model's

training dynamics (Figure 5). As observed in the loss evolution, the training loss exhibits periodic dips and spikes, which is entirely consistent with the Cosine Annealing learning rate decay and warm restarts actively helping the model escape local minima. However, the validation loss behaves differently: after an initial decrease during the first 50 to 100 epochs, it flattens out and progressively diverges from the training loss, indicating the onset of overfitting.

Interestingly, despite this clear divergence in the validation loss (which is based on Binary Cross-Entropy), the ranking metrics—AUROC and AUCPR—rise rapidly in the early epochs and subsequently maintain a highly stable plateau. This crucial discrepancy reveals that while the model's raw probability predictions gradually lose their absolute calibration (causing the loss to rise), its underlying ability to correctly *rank* positive and negative molecules remains firmly intact.

This behavior perfectly justifies our post-processing methodology. The drifting probability calibration explains why the naive approach utilizing a static 0.5 threshold performed poorly, trapping the F1-score at a low plateau around 0.22 during training. Because the robust ranking capability is preserved despite the diverging loss, extracting the optimal decision thresholds independently after training effectively rescues the model's utility. This step is what successfully translates the MPNN's stable AUROC into our state-of-the-art final F1-Score of 0.3895.

5.4. GATv2

To further investigate how dynamic feature weighting influences odor prediction, we expanded beyond the literature's baseline by implementing the Graph Attention Network v2 (GATv2). The theoretical advantage lies in its multi-head attention mechanism, which allows the model to dynamically assign importance weights to different neighboring atoms.

5.4.1. Training Dynamics and Metric Analysis

An analysis of the GATv2 learning curves reveals a consistent divergence between training and validation behavior across settings. In both the naive and optimized setups, the training loss steadily decreases throughout the epochs, indicating that the model fits the training data effectively. However, the validation loss decreases only during the initial phase and then gradually increases, revealing a clear overfitting pattern. This effect is noticeably more controlled in the stratified setting, where the validation curve is smoother and rises more slowly, suggesting that second-order stratification helps stabilize generalization.

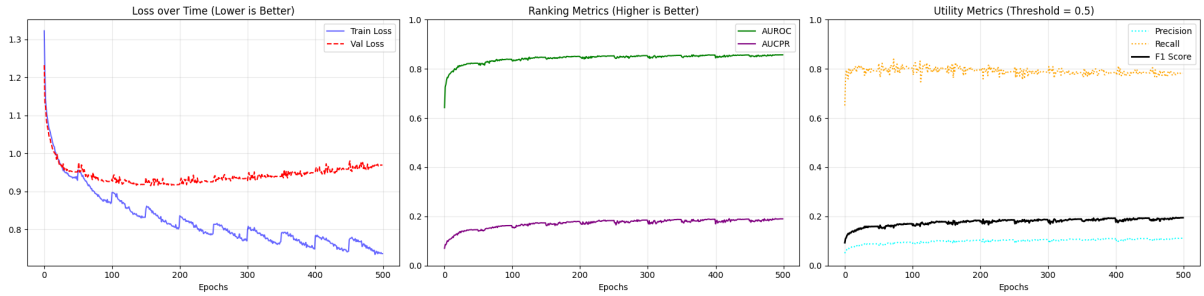


Figure 6. Training dynamics, ranking metrics, and naive F1-score evolution for the GATv2.

The ranking metrics (AUROC and AUPRC) further reinforce this observation. In both cases, AUROC increases rapidly in the early epochs and then plateaus, which is typical for

well-learning models. However, the stratified setup achieves a higher final AUROC ($\sim$0.856 vs $\sim$0.831), indicating better overall ranking capability. The AUPRC, which is highly sensitive to class imbalance, also benefits significantly from stratification, showing consistently higher values across epochs.

The most significant differences appear in the utility metrics. Without threshold optimization, the model exhibits very high recall ($\sim$0.71) but extremely low precision ($\sim$0.07), indicating that it predicts a large number of false positives. As a result, the F1 score remains very low ($\sim$0.13). With the introduction of optimal thresholding and stratification, the model achieves a much more balanced trade-off. Precision increases significantly ($\sim$0.24), while recall decreases to a more controlled level ($\sim$0.50), leading to a substantial improvement in the F1 score ($\sim$0.33).

Overall, these results demonstrate that while the underlying GATv2 model is highly capable of learning meaningful representations (achieving an AUROC identical to our optimized GCN), its practical performance in a multi-label setting is entirely dependent on post-processing choices. Stratification improves generalization and metric stability, while optimal thresholding is essential for translating good ranking performance into usable classification results.

6. Conclusion

Overall, this project confirms that predicting olfactory perception from molecular structure is a highly complex task that relies heavily on both advanced graph-based representations and rigorous data handling. While baseline and attention-based networks provide solid predictive capabilities, integrating edge-conditioned message passing architectures alongside second-order iterative stratification and independent threshold optimization allows the system to rival and exceed state-of-the-art metrics, effectively bridging the gap between chemical topology and biological perception. The next phase of this research could rely on applying this optimized pipeline to novel, independent datasets and exploring multimodal deep learning extensions.

7. Future Works

While the current methodology has successfully established a highly optimized pipeline for molecular odor prediction, there remain several promising avenues for future research. Our primary immediate objective is to evaluate the generalization capability of our models by training and testing them on a completely new, independent dataset. This dataset, utilized by a collaborating researcher associated with Professor Renton, presents a valuable opportunity to verify whether the structural representations and threshold calibrations learned by our MPNN and GCN architectures remain robust across fresh, unseen chemical samples.

Additionally, inspired by recent advancements in the state-of-the-art literature, future iterations of this project could explore multimodal feature integration. By ensembling our robust graph-based topological embeddings with complementary features, such as advanced physicochemical descriptors or visual CNN embeddings, the aim could be to capture an even wider spectrum of the complex structure-odor relationship and push the predictive metrics even higher.

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